

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Metallurgical and Materials Engineering
2. Subject Name: Mechanical testing and working of materials, L-T-P: 3:0:0, Credits:3
3. Pre-requisites: None
4. Syllabus and reference books:

Syllabus:

Mechanical testing: Hardness – Brinell, Rockwell, Vickers, Micro-Vickers, Nanoindentation; Tensile– Engineering and true stress-strain curves, Elastic modulus, yield and ultimate tensile strengths, ductility; Compression– Influence of aspect ratio, lubricants; Flexural and torsion; Creep – Stages and mechanisms of creep, Creep strain rate, Creep rupture strength; Impact – Ductile to Brittle Transition temperature; Fracture Toughness – Basics of Fracture Mechanics, Griffith's Criteria, concept of plastic zone size, Plane strain and Plane stress conditions; Fatigue – Stages of Fatigue, High Cycle Fatigue, Low Cycle Fatigue, Fatigue Crack Growth Rate testing.

Mechanical working: Fundamentals of Metal working, Classification of forming processes, Mechanics of Metalworking – Slab method and Uniform energy deformation method, Flow stress determination, Temperature in metal working - Hot-working and Cold-working, Strain rate effects, Microstructure evolution, Texture evolution, Forging, Friction and lubrication, Ring compression tests, Rolling, Extrusion, Wire and Tube drawing, Residual stress, Sheet metal forming, Instability and flow localization, Workability, High energy rate forming.

Reference Books:

- 1) G. E. Dieter, Mechanical Metallurgy, McGraw Hill Book Company.
- 2) A. Bhaduri, Mechanical properties and Working of Metals and Alloys, Springer Series in Materials Science 264.
- 3) J.N. Harris: Mechanical Working of Metals- Theory and Practice, Pergamon Press, Oxford.
- 4) Deformation and Fracture Mechanics of Engineering Materials by Richard Hertzberg, Richard P. Vinci and Jason L. Hertzberg, John Wiley and Sons publications.

5. Lecture-wise break-up:

S. No.	Topic	No. of lecture hours
1.	Mechanical testing of Materials:	
1.1.	Hardness testing-Brinell, Rockwell, Vickers, Micro-Vickers, Nanoindentation	2

1.2.	Tension testing – Engineering and true stress-strain curve generation, Elastic modulus, yield and ultimate tensile strengths, ductility	2
1.3.	Compression Testing – Influence of aspect ratio, lubricants	1
1.4.	Flexural testing (torsion)	1
1.5.	Creep – Stages and mechanisms of creep, Creep strain rate, Creep rupture strength	2
1.6.	Impact- Ductile to Brittle Transition temperature, Factors affecting DBTT	2
1.7.	Fracture-Griffith's Criteria, concept of plastic zone, Plane strain and Plane stress	3
1.8	Fatigue – Stages of Fatigue, High Cycle Fatigue, Low Cycle Fatigue, Fatigue Crack Growth Rate	2
2.	Mechanical working of materials	
2.1.	Fundamentals of Metal working, Classification of forming processes.	1
2.2.	Mechanics of Metalworking: Slab method and Uniform energy deformation method	2
2.3.	Flow stress determination for metal working, Temperature in metal working - Hot-working and Cold-working, Annealing, Strain rate effects,	2
2.4.	Microstructure evolution, (microstructural hierarchy – cells/subgrains, cell bands or blocks, Deformation and transition bands, micro-shear bands, Shear banding and flow lines formation, Effect of stacking fault energy, twins, dynamic recovery and recrystallization mechanisms and behavior, formation of statistically stored dislocations, incidental dislocation boundaries, and geometrically necessary boundaries.	3
2.5.	Texture evolution (Deformation behavior in single crystal – primary slip and conjugate slip, deformation behavior in polycrystalline material: Slip and twinning, Crystal plasticity, Schmid's law, Taylor model, Visco-plastic self consistent modelling)	2
2.6.	Introduction to forging operations, friction and lubrication, ring compression tests, Rolling, Extrusion, Wire and Tube drawing processes, Residual stress.	3
2.7.	Introduction to sheet metal forming, Instability and flow localization, Workability, High energy rate forming processes.	2
	Total number of hours	30